

Synoptic CB: Porewater Nutrients

November 2025 Samples

2026-08-21

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##Run Information

```
cat("Run Information: NAME ") #lets you know what section you're in
```

Run Information: NAME

```
#set the run date & user name
```

```
run_date <- "07/01/2026"  
sample_year <- "2025"  
sample_month <- "NOVEMBER"  
user <- "Stephanie Wilson"
```

```
##Data entered incorrectly
```

```
Old_ID_1 <- "SWH_202505_SWMAP_LysB_45cm"  
New_ID_1 <- "SWH_202505_SWAMP_LysB_45cm"
```

```
#identify the files you want to read in
```

```
#read in as a list to accommodate multiple runs in a month
```

```
NOx_files <- c("Raw Data/SEAL_COMPASS_Synoptic_NOx_202511_1.csv",  
              "Raw Data/SEAL_COMPASS_Synoptic_NOx_202511_2_GOOD.csv") #,  
              #"Raw Data/SEAL_COMPASS_Synoptic_NOx_202511_3.csv")  
NH3_PO4_files <- c("Raw Data/SEAL_COMPASS_Synoptic_NH3_PO4_202511_1.csv",  
                  "Raw Data/SEAL_COMPASS_Synoptic_NH3_PO4_202511_2.csv")  
              # "Raw Data/SEAL_COMPASS_Synoptic_NH3_PO4_202511_3.csv",  
              # "Raw Data/SEAL_COMPASS_Synoptic_PO4_202511_2_GOOD.csv")
```

```
# Define the file path for QAQC log file - NO Need to change just check year
```

```
file_path <- "Raw Data/SEAL_COMPASS_Synoptic_QAQC_Log_2025.csv"  
final_path <- "Processed Data/COMPASS_Synoptic_Nutrients_202511.csv"
```

```
#record any notes about the run or anything other info here:
```

```
run_notes <- "No NOx reduction efficiency tests run. It looks like they were forgotten, but the rest of the run was fine"
```

```
#Set up file path for metadata
```

```
#downloaded metadata csv - downloaded from Google drive as csv for this year  
Raw_Metadata = "Raw Data/COMPASS_SynopticCB_PW_SampleLog_2025.csv"
```

##Setup

##Read in metadata and create similar sample IDs for matching to samples

0.1 Import Data & Clean

```
##Fix data entered incorrectly
```

```
####Fix Sample IDs####
#df_all$Sample_Name[df_all$Sample_Name == Old_ID_1] <- New_ID_1

#move the date to after the site id:
df_all$Sample_Name <- sub(
  "^[0-9]{6})_(GCW|GWI|MSM|SWH)_(TR|WC|UP|SW)_(LysA|LysB|LysC)_(10|20|45|NA)cm$",
  "\\2_\\1_\\3_\\4_\\5cm",
  df_all$Sample_Name
)

#also change the SW ones
df_all$Sample_Name <- sub(
  "^[0-9]{6})_(GCW|GWI|MSM|SWH)_(SW)_(A|B|C)$",
  "\\2_\\1_\\3_\\4",
  df_all$Sample_Name
)
```

0.2 Assessing standard Curves

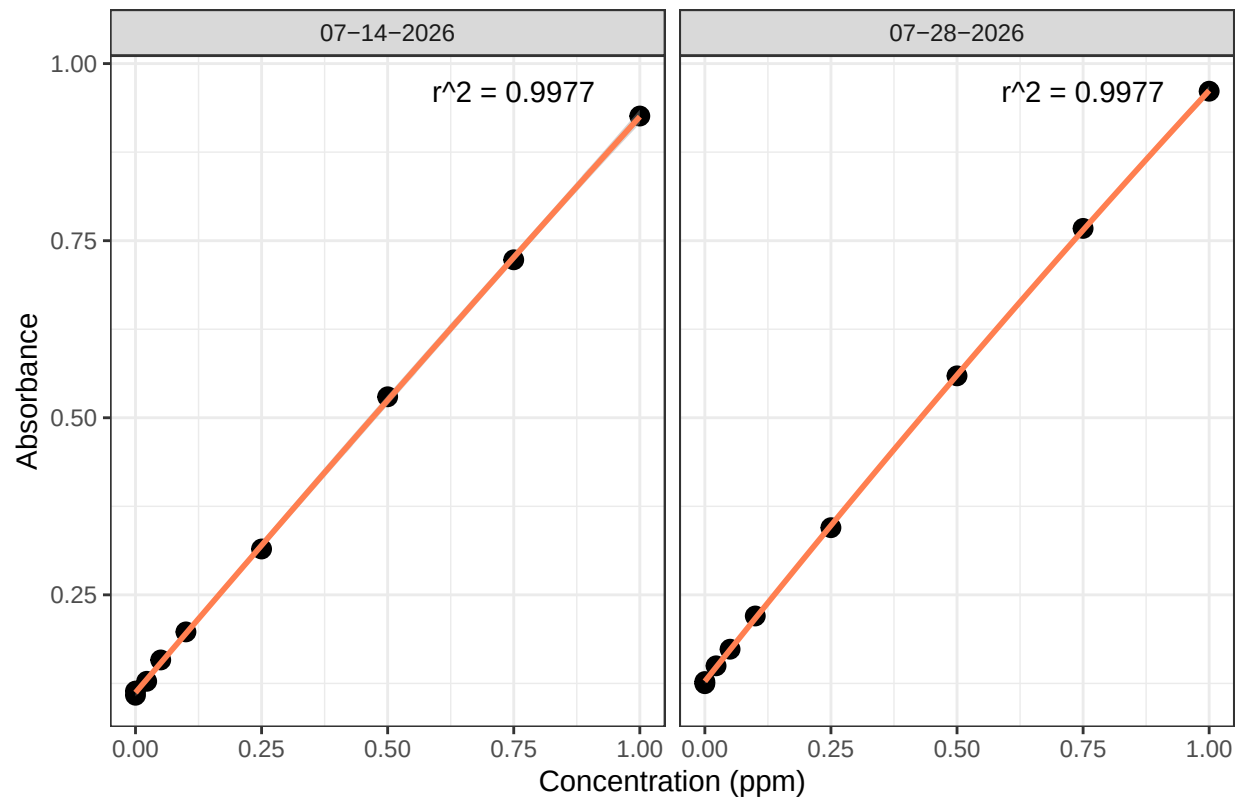
```
#Pull out standards data
```

```
## Assess Standard Curves
```

0.3 Plot standards data

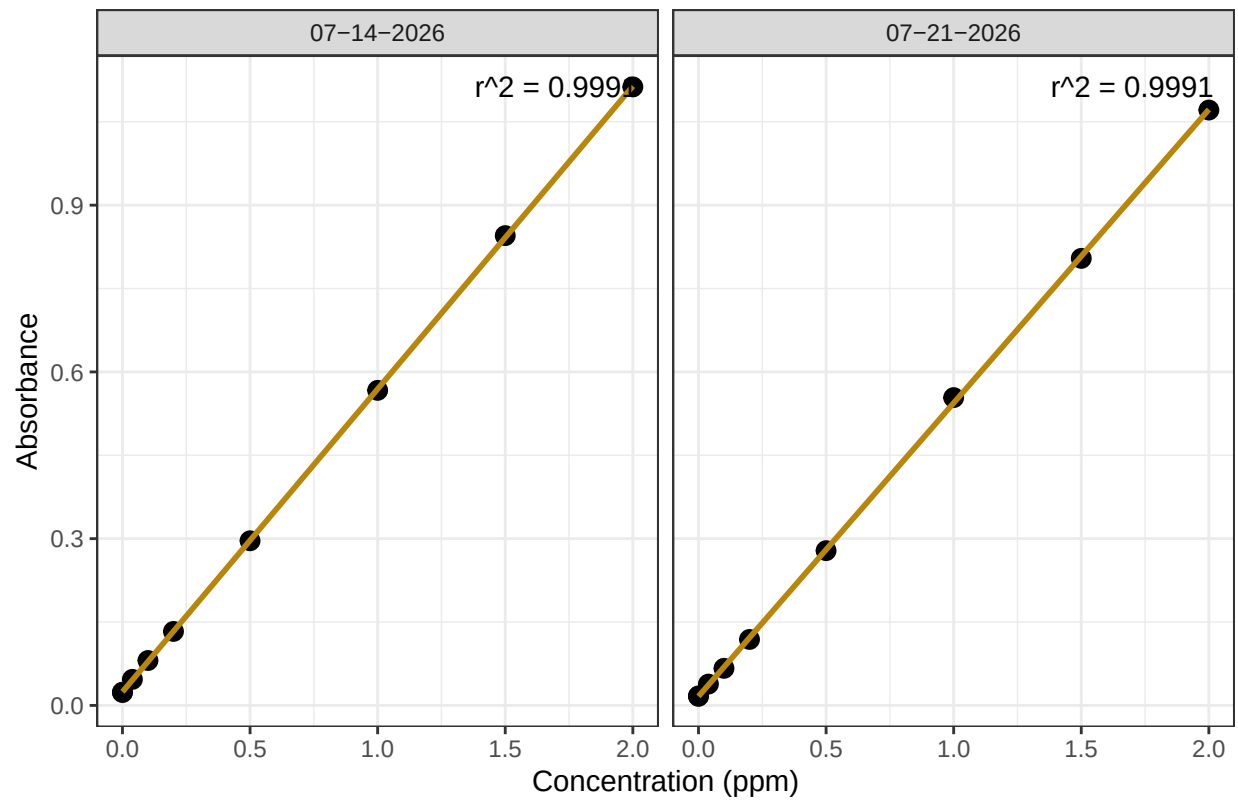
```
## Assess Standard Curves
```

NOx Standard Curve



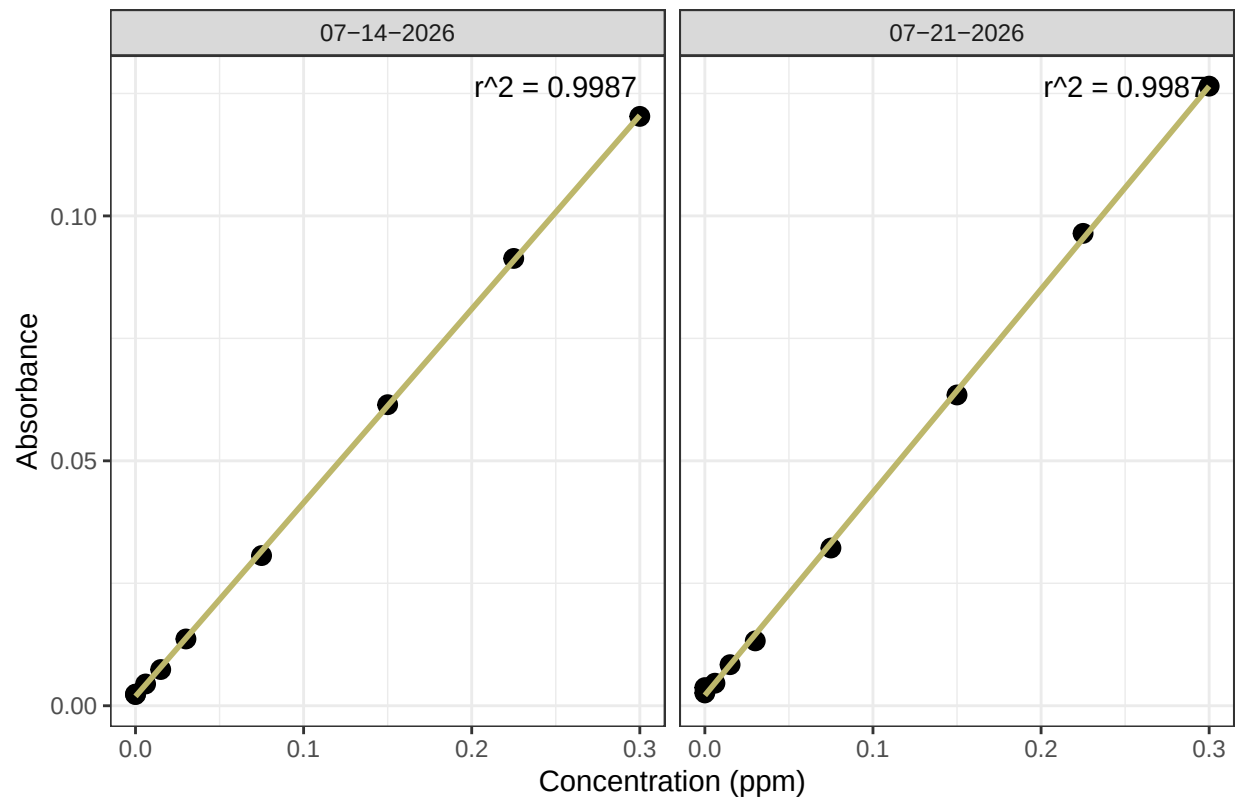
```
## 'geom_smooth()' using formula = 'y ~ x'
```

NH3 Standard Curve



```
## 'geom_smooth()' using formula = 'y ~ x'
```

PO4 Standard Curve



```
## [1] "NOx Curve r2 GOOD - PROCEED"
```

```
## [1] "NH3 Curve r2 GOOD - PROCEED"
```

```
## [1] "PO4 Curve r2 GOOD - PROCEED"
```

```
## [1] "QAQC log file exists and has been read into the code."
```

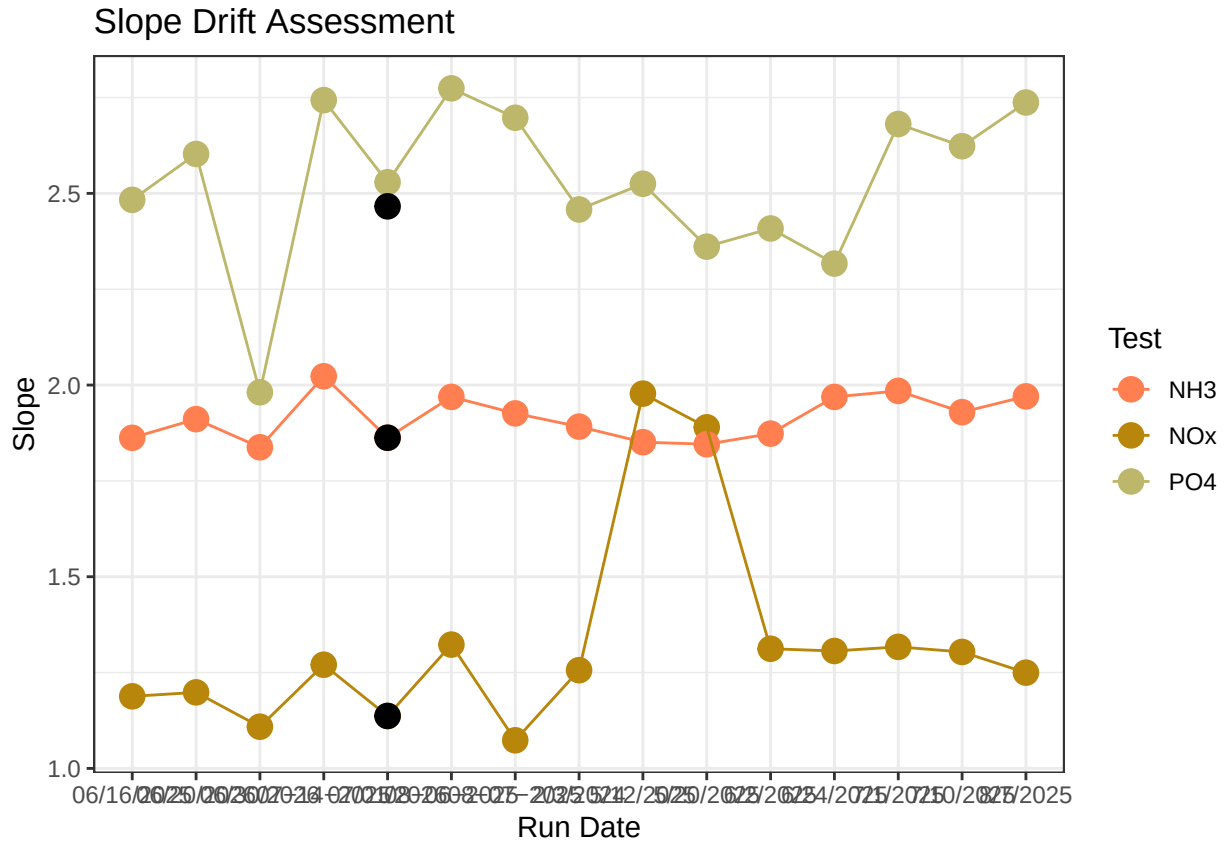


Table 1: Average Slope by Analyte

Test	avg_slope
NH3	1.914
NOx	1.327
PO4	2.528

0.4 Dilution Corrections - ensure the latest dilution is kept

```
## Dilution Corrections
```

```
## Duplicated samples: SWH_202511_TR_LysA_45cm, SWH_202511_TR_LysB_10cm, SWH_202511_TR_LysB_20cm, SWH_202511_TR_LysB_45cm
```

```
##
```

```
## All duplicated samples have valid dilutions. No naming issues detected.
```

0.5 Performance Check

```
## [1] "NOx pe Check has a % Difference >25% - REASSESS"
```

```
## Run mean = 3.579636
```

```
## Expected = 1.51
```

```
## [1] "NH3 pe Check has a % Difference <25% - PROCEED"
```

```
## Run mean = 1.246927
```

```
## Expected = 1.034
```

```
## [1] "PO4 pe Check has a % Difference <25% - PROCEED"
```

```
## Run mean = 0.862585
```

```
## Expected = 0.824
```

```
##Check NOx Reduction Efficiency - No Reduction Efficiencies Run - forgotten
```

```
“{#r Assess reduction Efficiency, echo=FALSE} cat(“Assess Reduction Efficiency”)
```

```
#check on the reduction efficiency for the NOx test: #Pull out test stds red_eff <- df_all %>% filter(Sample_Name %in% c(“Nitrate Standard”, “Nitrite Standard”))
```

```
red_effEffPercent <- (red_effConc / 0.5)*100 red_effEffFlag <- ifelse(red_effEffPercent >= 90, ‘YES’, ‘NO, rerun’)
```

```
red_eff <- red_eff %>% mutate(row_num = row_number())
```

```
#Plot these efficiencies red_effs_plot <- ggplot(data = red_eff, aes(x = factor(row_num), y = Conc, fill=Eff_flag)) + geom_bar(stat = ‘identity’) + facet_wrap(~Sample_Name) + scale_fill_manual(values = c(“YES” = “midnightblue”, “NO, rerun” = “darkred”)) + theme_classic() + labs(x= “”, y=“NOx (mg/L)”, title=“”) + theme(legend.position=“bottom”) + geom_hline(yintercept=0.5, linetype=“dashed”, color = “black”, linewidth=1) + guides(fill=guide_legend(title=“Reduction Efficiency >90%”))+ scale_x_discrete(drop = TRUE) print(red_effs_plot)
```

```
#create a flag NOx_mean_red_eff <- mean(red_eff$Eff_Percent)
```

```
#report out if flags indicate need for rerun and what the value is ifelse(NOx_mean_red_eff >= 95, “Mean NOx Reduction Efficiency >95% - PROCEED”, “Mean NOx Reduction Efficiency <95% - REASSESS”) print(NOx_mean_red_eff)
```

```
#write out a flag to the sample dataframe if red eff is low if (NOx_mean_red_eff <= 95) { df_allNOxFlag <- ifelse(df_allNOxFlag != “”, paste0(df_all$NOx_flag, “; NOx reduction efficiency low”), “NOx reduction efficiency low”)} }
```

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Analyze the Check Standards

0.6 Analyze Check Standards

0.7 [1] “NO_x Check Standard RSD within Range - PROCEED”

0.8 [1] “NH₃ Check Standard RSD within Range - PROCEED”

0.9 [1] “PO₄ Check Standard RSD within Range - PROCEED”

![] (COMPASS_Synoptic_NUTR_Analysis_202511_files/figure-latex/Check Standards-1.pdf)<!-- -->

0.10 [1] “>60% of NO_x Check Standards are within range of expected concentration - PROCEED”

0.11 [1] “>60% of NH₃ Check Standards are within range of expected concentration - PROCEED”

0.12 [1] “>60% of PO₄ Check Standards are within range of expected concentration - PROCEED”

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Analyze Blanks

0.13 Assess Blanks

0.14 [1] “>60% of NOx Blanks are below the lower 25% quartile of samples or 1/2 detection limit - PROCEED”

0.15 [1] “>60% of NH3 Blanks are below the lower 25% quartile of samples - PROCEED”

0.16 [1] “>60% of PO4 Blanks are below the lower 25% quartile of samples- PROCEED”

<!-- -->

Table: Mean Concentration of Blanks

Test	Blank_Mean_Conc
NOx	0.1242
NH3	0.0013
PO4	0.0014

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Analyze Duplicates

0.17 Analyze Duplicates

0.18 [1] “>60% of NOx Duplicates have a CV <10% - PROCEED”

0.19 [1] “>60% of NH3 Duplicates have a CV <10% - PROCEED”

0.20 [1] “<60% of PO4 Duplicates have a CV <10% - REASSESS”

<!-- -->

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Spikes

0.21 [1] “>60% of Spikes have a CV <50% - PROCEED”

0.22 [1] “>60% of Spikes have a CV <50% - PROCEED”

0.23 [1] “>60% of Spikes have a CV <50% - PROCEED”

0.24 Warning: Using size aesthetic for lines was deprecated in ggplot2 3.4.0.

0.25 i Please use linewidth instead.

0.26 This warning is displayed once per session.

0.27 Call lifecycle::last_lifecycle_warnings() to see where this warning was

0.28 generated.

```
![] (COMPASS_Synoptic_NUTR_Analysis_202511_files/figure-latex/Analyze Spikes-1.pdf)<!-- -->
```

```
\newpage
```

```
## Matrix Effects
```

0.29 [1] “NO NOx Matrix Effect, PROCEED”

0.30 [1] “NO NH3 Matrix Effect, PROCEED”

0.31 [1] “NO PO4 Matrix Effect, PROCEED”

```
## Unit Converted Data Column Added (mg/L to uM )
```

```
## Sample Flagging - Within range of standard curve
```

0.32 Sample Flagging

```
## Pull out sample id information
```

0.33 Sample Processing

```
## Check to see if samples run match metadata & merge info
```

0.34 Check Sample IDs with Metadata

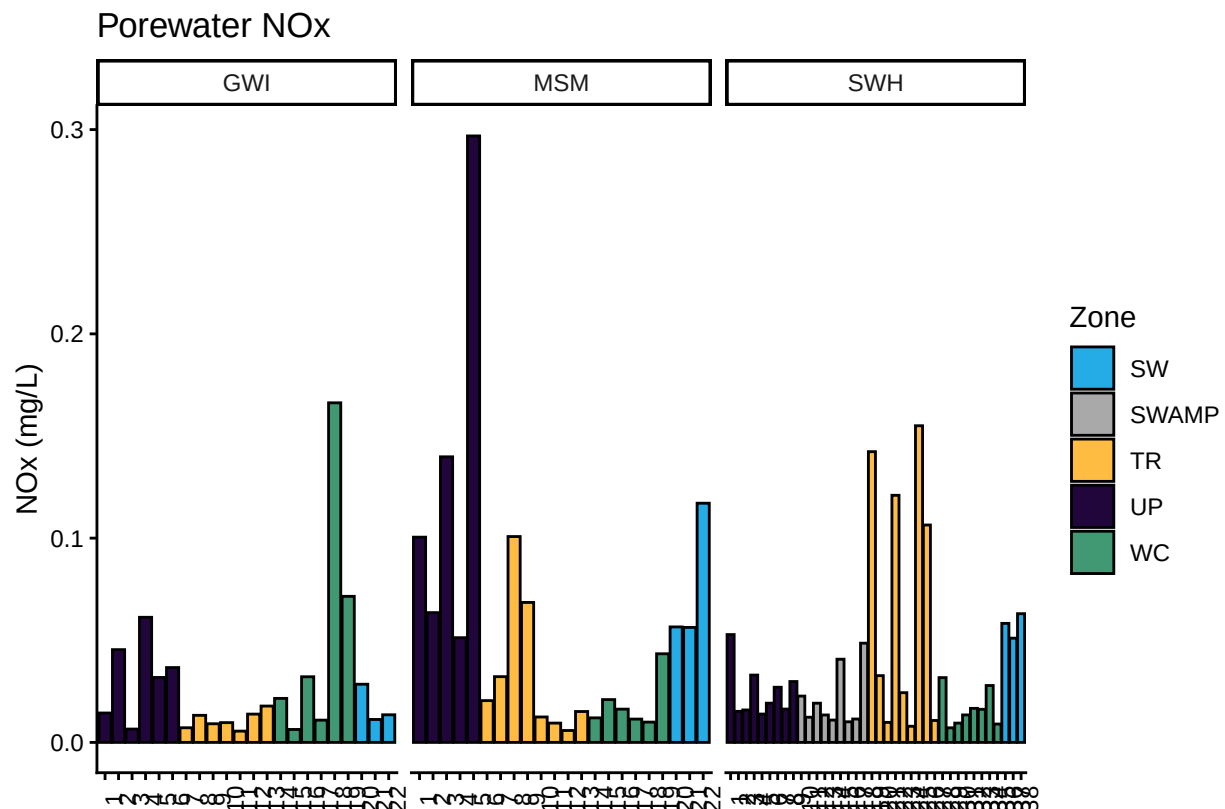
0.35 All sample IDs are present in metadata.

```
\newpage
```

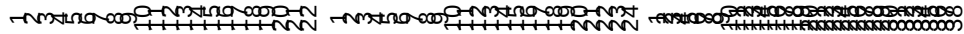
```
## Visualize Data
```

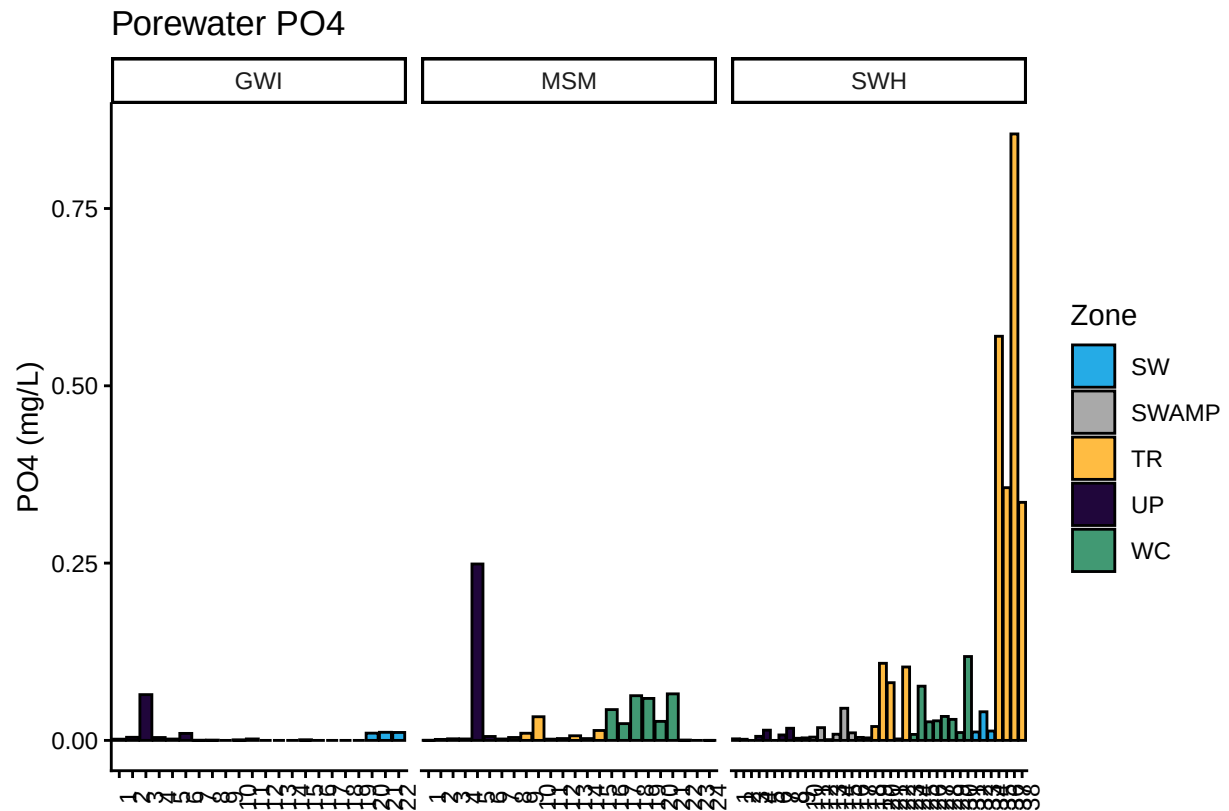
0.36 Visualize Data

```
““
```



GWI	MSM	SWH
-----	-----	-----





0.37 Export Processed Data

#end